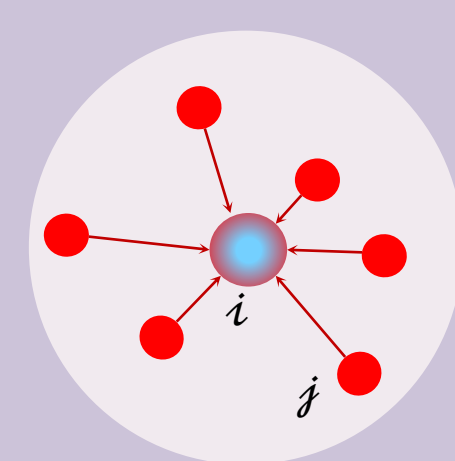


The Language of Crystals

Conditional generation for crystal structures

Where do the conditions go?



```
data_TiO2
_symmetry_space_group_*
'P4_2/mnm'
_cell_length_a 4.6533
(...)
O O1 4 0.1954 0.8046 0.5000 1
```

- Words only make sense in context
- Atoms only make sense in the context of their environment
- A transformer learns by repeatedly asking: *given everything written so far, which chunk should come next?*

$$p_{\theta}(x_t | x_{<t})$$

$$p_{\theta}(\text{CIF}) \rightarrow$$

Brookite?
Anatase?
Rutile?

$$p_{\theta}(\text{CIF} | c_{\text{XRD}}) \rightarrow$$

Rutile

Prompting with TiO_2 only lets us resolve the polymorph if additional characterisation is given

- We can extend the language model to not only produce a valid CIF
- But also, to respect a particular property or target e.g. good PV, has this XRD profile – any numerical target goes

Suited for natural language

```
<XRD_peak_1> 25.1 100
<XRD_peak_2> 37.8 35
data_TiO2
(...)
O O1 4 0.1954 0.8046 0.5000 1
```

Digits split into tokens

Heterogeneous conditions
(missing peaks etc.)
destabilise training

Same token space for
properties + CIF syntax

Suited for chemistry

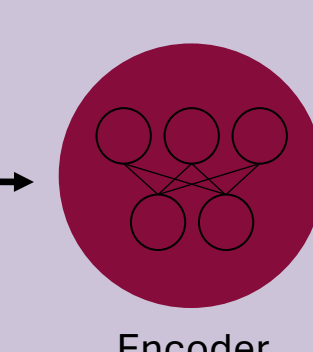
```
data_TiO2
(...)
O O1 4 0.1954 0.8046 0.5000 1
```

Conditions enter as
continuous signals

CIFs unaltered

Pretrained crystal
knowledge is preserved
during property fine-tuning

$$c_{\text{XRD}} = [(2\theta_1, I_1), \dots, (2\theta_N, I_N)]$$



Encoder

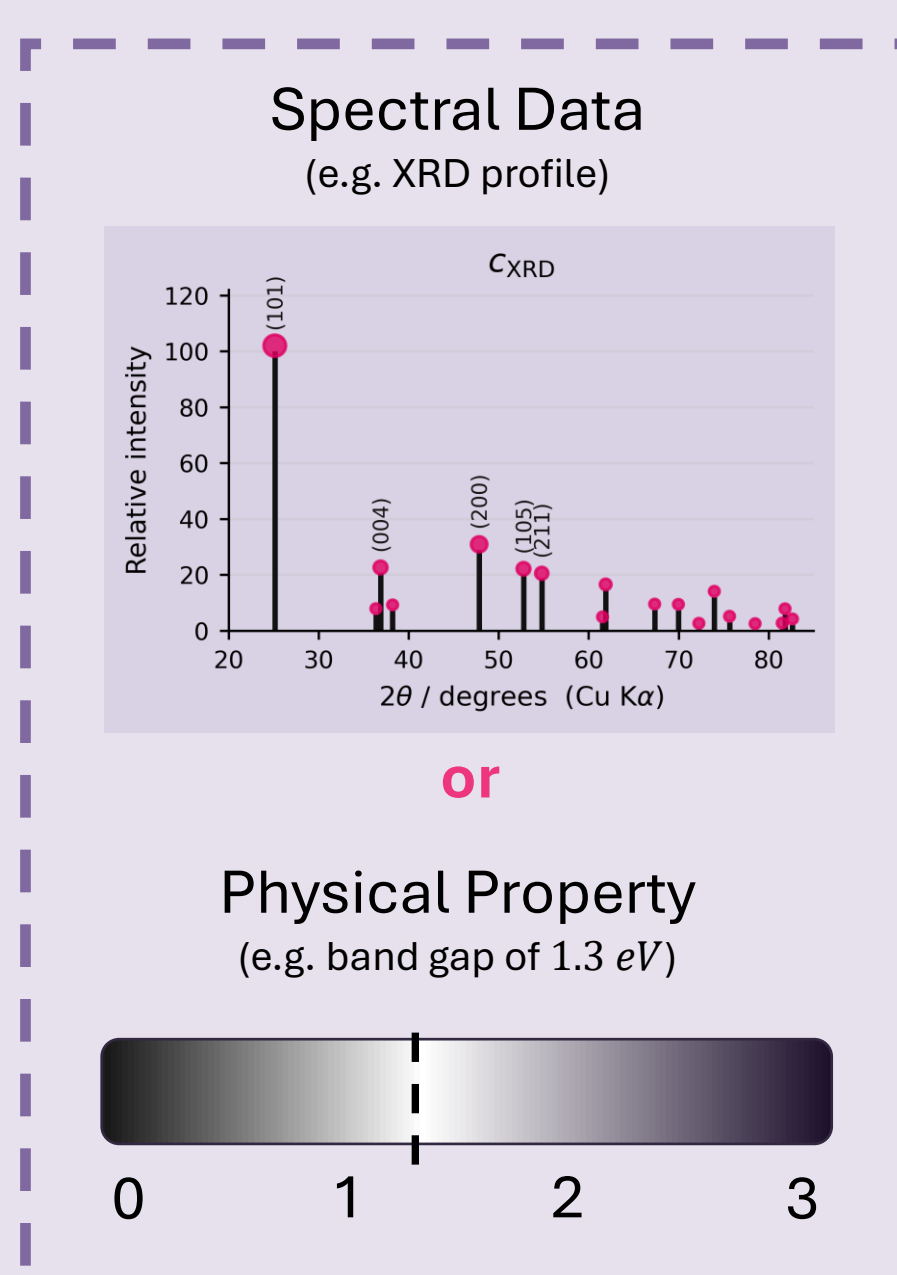
Framework

III. CrystaLLM-π Transformer

IV. Output

V. Validate

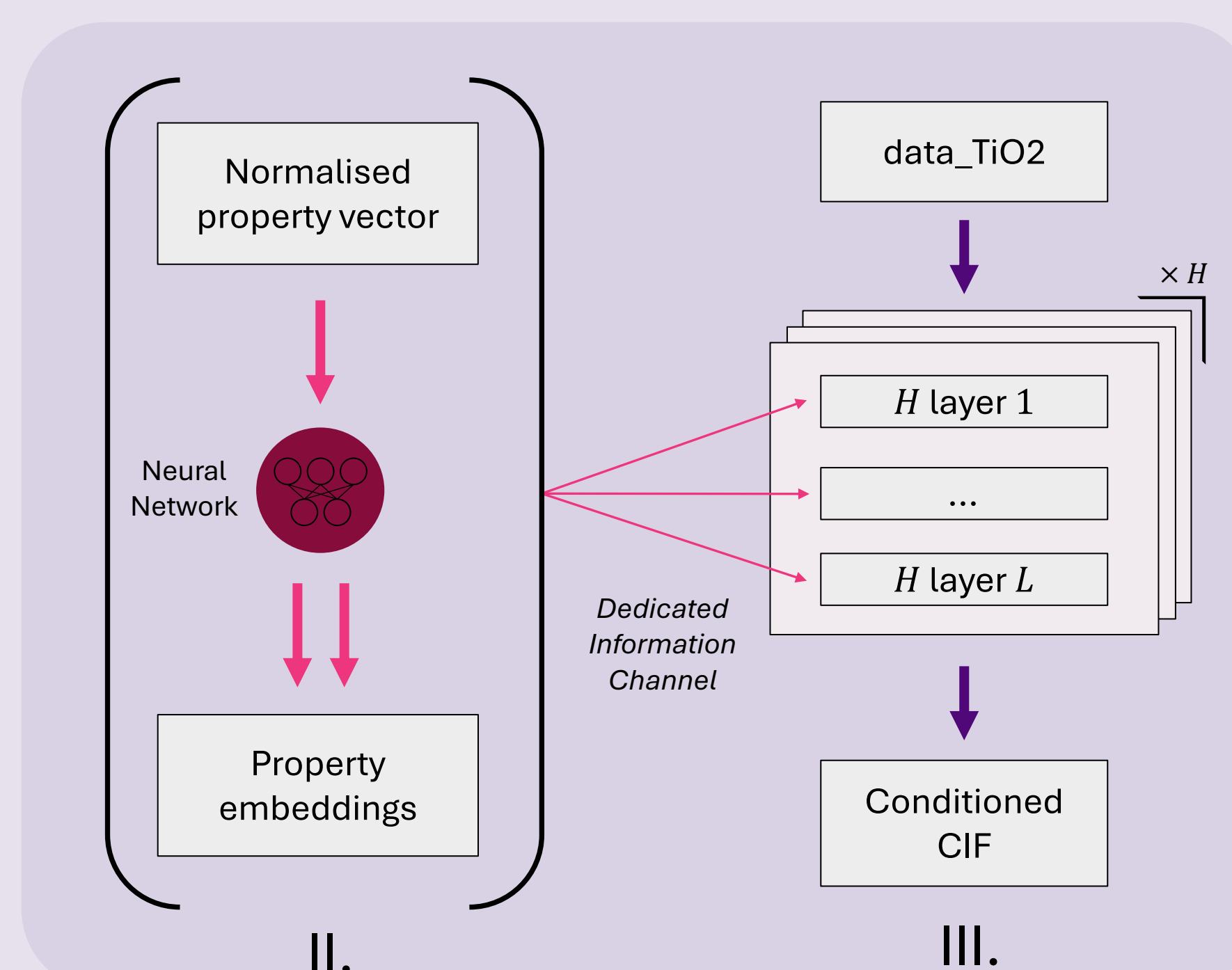
I. Target Property



II. Continuous transformation of property

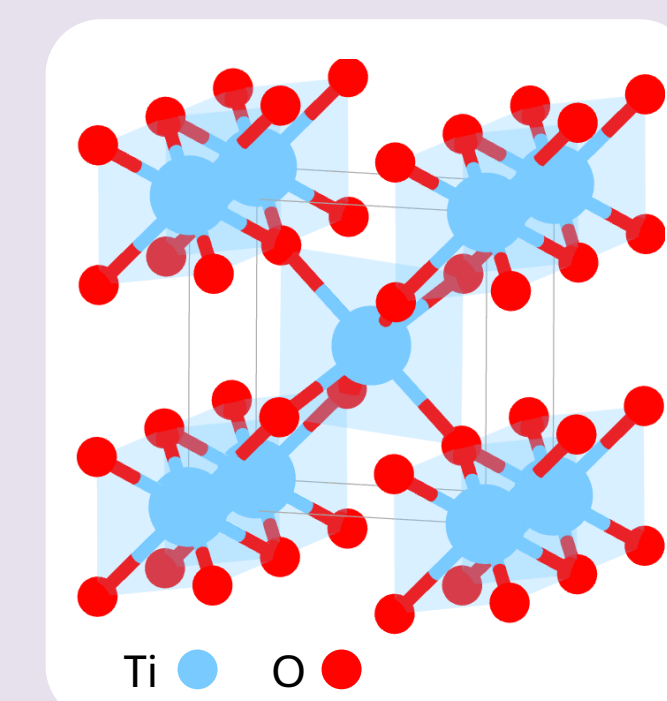
Properties injected at every decoding step

$$p_{\theta}(\text{CIF} | c_{\text{XRD}}) = \prod_{t=1}^T p_{\theta}(x_t | x_{<t}, c_{\text{XRD}})$$



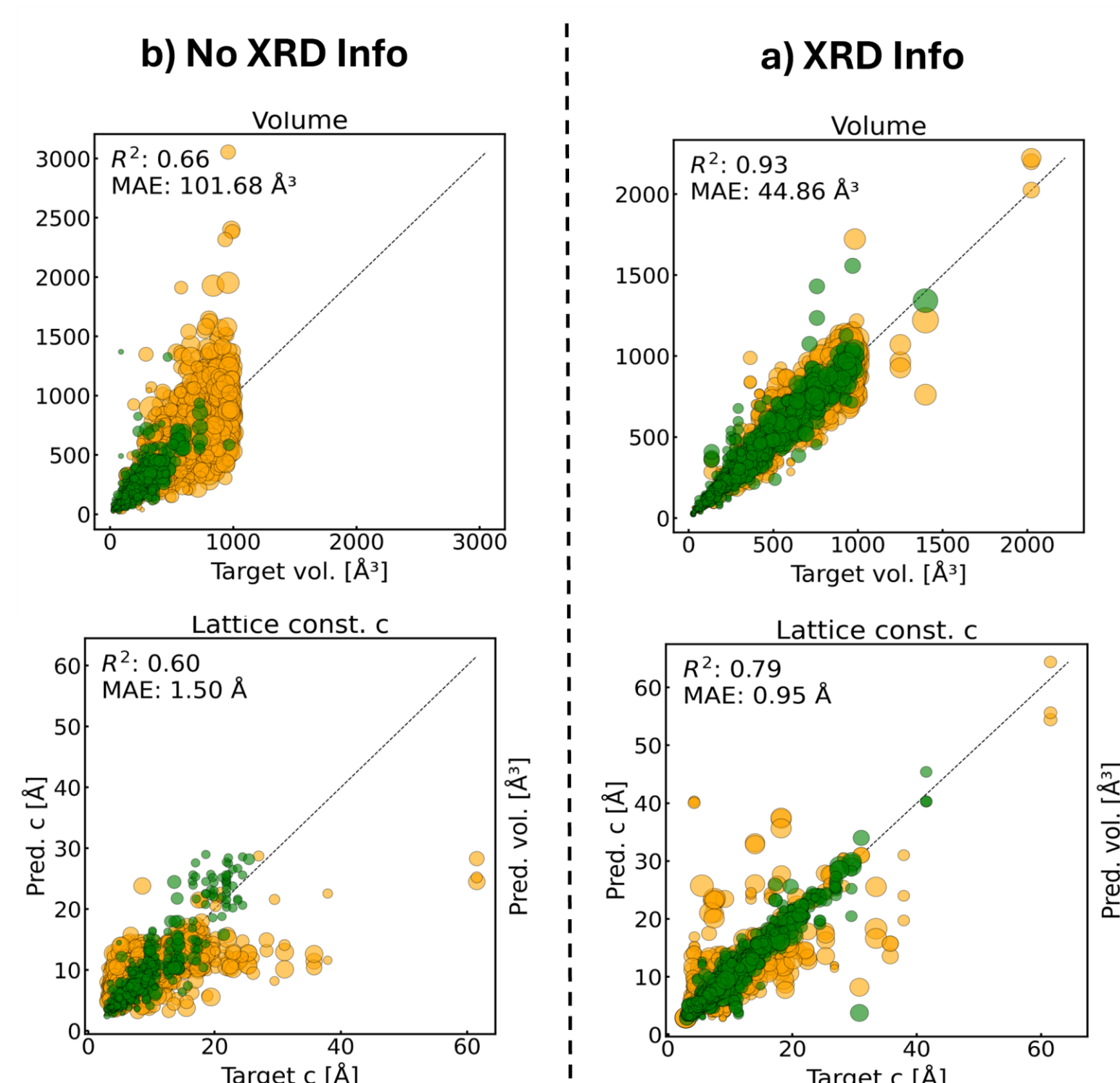
Pre-trained on 4 million unlabelled CIFs

```
data_TiO2
_symmetry_space_group_*
'P4_2/mnm'
_cell_length_a 4.6533
(...)
O O1 4 0.1954 0.8046 0.5000 1
```



- ☒ Composition
- ☒ Symmetry
- ☒ Bond Lengths
- ☒ Pred. Stability
- ☒ Band Gap
- ☒ Novelty
- ☒ DFT Checks

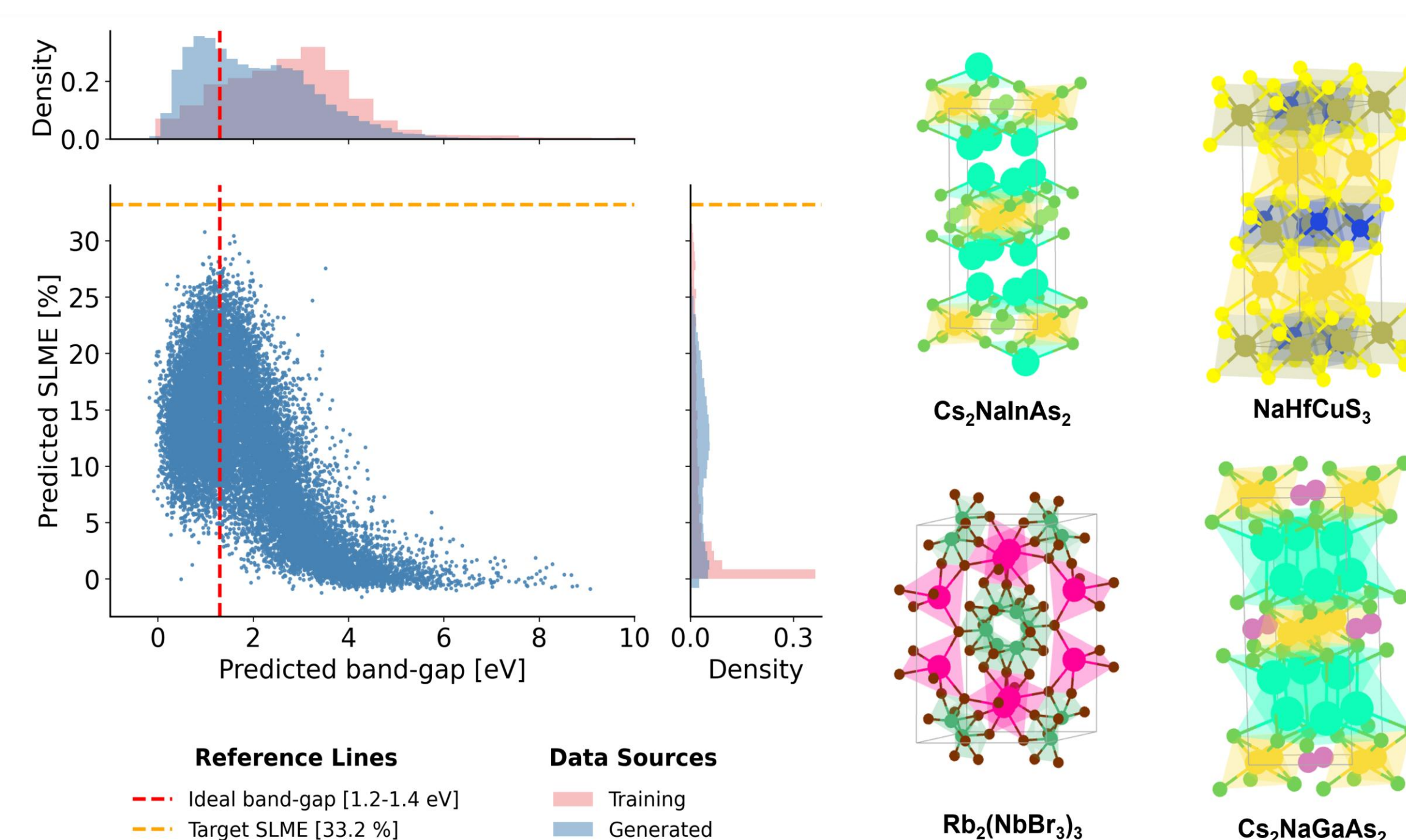
Experimental structure recovery



Predicted vs. True lattice parameter (c-axis) and Volume for generated experimental structures

- Evaluated on a test set of 1.5K structures from the Crystallographic Open Database (COD)
- XRD conditioning improves structure-match rate from **15.89%** to **49.04%**
- Conditioned model recovers structures with up to **83 atoms per unit cell** vs. **40** for unconditioned

Photovoltaic Material Discovery

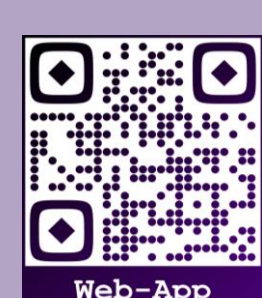


Distribution of the generated materials' properties vs training set (left). Promising candidates generated (right)

- Fine-tuned on 5.4K structures with labelled Spectroscopic Limited Maximum Efficiencies (SLMEs)
- Generated candidates cluster near optimal single-junction Shockley-Queisser limit band gap range (1.2 – 1.4 eV)
- Compounds absent from training data generated, with SLMEs & stability validated by DFT (HSE06 functional used)

What Next?

- Open-Source Web-App:** Try our interactive generation tool, with XRD conditioning support & automatic composition Z-search.
- Future Directions:** Improving XRD conditioning to avoid peak-pick pre-processing, modelling larger or disordered structures...



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- API and Web-app developed in partnership with Physical Sciences Data Infrastructure (PSDI)

